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Potential benefits and challenges of artificial intelligence in human resource management in public institutions

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Abstract
This study explores the benefits and challenges of implementing artificial intelligence (AI) in human resource management (HRM) among public institutions in Tanzania. Using a cross-sectional research design, data was collected from 217 HR practitioners through random sampling and questionnaires. The descriptive approach assessed participants' understanding of AI, its perceived benefits, challenges, and risks in HRM. The findings on the perceived benefits of AI in HRM show that increased efficiency, better decision-making, and cost reduction are the top advantages, while ease of use ranks the lowest. However, challenges such as lack of expertise, data privacy concerns, high costs, and resistance to change remain significant barriers. Most respondents strongly agree that AI enhances recruitment, training, performance management, and compliance, but concerns persist over bias, transparency, and emotional intelligence limitations in areas like employee relations and compensation. Additionally, results on risks of using AI on HRM indicate that high-risk components like Human Resource Information System (HRIS), Recruitment, Compensation, and Performance Management face serious concerns such as bias and implementation complexity. Moderate-risk areas like succession planning and compliance show manageable challenges, while low-risk components like health and safety and employee relations reflect minimal issues. Statistical analysis confirms significant associations between risk levels and several HRM components, especially HRIS and Recruitment ($p < 0.01$), highlighting the need for targeted interventions. The study emphasizes the need for public institutions to establish robust ethical frameworks, invest in capacity-building for HR practitioners, and ensure human oversight in AI-driven processes. By addressing these concerns, public institutions can mitigate risks, improve AI adoption, and optimize the potential of AI to enhance HR practices, offering a roadmap for equitable, effective, and ethical implementation of AI technologies in workforce management.

Keywords OpenAI · Digitalization · Smart HR · AI-powered recruitment · ChatGPT

1 Introduction

The strategic management of human resources has become a focal point for organizations globally, as they seek to enhance both individual and institutional performance [1, 2]. Human Resource Management (HRM) has progressively evolved from administrative functions to a more strategic, technology-enabled domain [2–4]. Among the most transformative advancements in this space is the adoption of Electronic Human Resource Management (e-HRM), which leverages digital tools to improve HR processes such as recruitment, performance appraisal, training, and employee engagement

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[5, 6]. E-HRM has been lauded for its ability to reduce costs, improve decision-making, and streamline workflows, thereby enhancing organizational efficiency and sustainability [7, 8].

In recent years, Artificial Intelligence (AI) has emerged as a significant technological disruptor with profound implications for HRM [9]. Broadly defined, AI refers to the simulation of human intelligence in machines that are programmed to think, learn, and problem-solve autonomously or semi-autonomously [9, 10]. In the context of HRM, AI encompasses a range of applications, including talent acquisition through predictive analytics, employee sentiment analysis, chatbot-enabled HR support, and automation of routine administrative tasks [11, 12]. These AI-driven innovations promise to improve accuracy, reduce bias, increase responsiveness, and free up HR professionals to focus on strategic issues [13]. However, while the theoretical benefits of AI are well-documented, its actual implementation and impact remain uneven—particularly in developing countries [14].

The adoption of AI in HRM is influenced by a variety of models and frameworks that consider technological, organizational, and environmental factors [15]. For instance, the Technology-Organization-Environment (TOE) framework and the Diffusion of Innovations (DOI) theory are frequently used to assess organizational readiness, perceived usefulness, ease of use, and external pressures that drive or hinder adoption [14–16]. These models emphasize that successful AI integration requires not only technological infrastructure but also leadership support, workforce capabilities, regulatory alignment, and cultural adaptability.

Yet, much of the existing research and application of AI in HRM is centered in developed countries, leading to a lack of contextualized understanding in low- and middle-income regions, including Africa [17, 18]. Within this context, Tanzania offers a compelling case. While AI technologies are gradually being introduced in sectors such as agriculture, finance, and education [19, 20], there is limited empirical evidence on their integration into HRM practices—particularly within the public sector. This is surprising given that Tanzania's public institutions are among the largest employers in the country and play a vital role in national development.

Tanzania's legal and institutional frameworks present both opportunities and constraints for AI adoption [21]. The National ICT Policy (2016) and subsequent regulatory guidelines acknowledge the transformative potential of AI and digital technologies, yet they lack specific provisions addressing AI governance, especially in HRM [21, 22]. Moreover, the public sector often faces structural challenges such as bureaucratic rigidity, limited digital infrastructure, and low digital literacy among employees, all of which can impede technological innovation [23, 24]. Compounding this are gaps in compliance mechanisms and data protection policies, which raise ethical and legal concerns around privacy, transparency, and fairness—especially when AI is used in personnel decision-making processes.

Despite these challenges, anecdotal evidence suggests that some public sector employees in Tanzania are informally integrating AI tools to improve individual productivity, particularly in administrative and record-keeping tasks [25, 26]. This underlines a significant research gap: the absence of systematic studies on how AI is being adopted, experienced, and regulated within Tanzanian public sector HRM. Addressing this gap is essential to understanding how AI can support or hinder strategic human resource functions, what barriers institutions face, and how policy can evolve to ensure equitable and effective AI integration. Therefore, this study seeks to fill that gap by exploring the benefits, challenges, and dynamics of AI adoption in HRM within Tanzania's public sector. By situating the research within established AI adoption frameworks and grounding it in the Tanzanian legal and institutional context, the study offers a nuanced exploration of the interplay between technology, governance, and human capital development. The findings aim to inform policymakers, human resource practitioners, and technology developers about how to optimize AI use in HRM, ensuring that it complements human capabilities, enhances service delivery, and aligns with national development goals.

1.1 Theoretical framework

This study is grounded in two widely recognized frameworks that explain how organizations adopt and implement new technologies: the Technology-Organization-Environment (TOE) framework and the Diffusion of Innovations (DOI) theory. These frameworks are particularly relevant for examining the integration of artificial intelligence (AI) in human resource management (HRM), especially in the public institutions in developing countries like Tanzania. Both theories emphasize that the decision to adopt technology is not solely based on its technical qualities but also influenced by a range of organizational and environmental factors [27]. While the TOE framework provides a structured view of the internal and external conditions that affect adoption [28], the DOI theory focuses on how innovations spread through a system and what attributes make them more likely to be embraced [29]. Together, these perspectives allow for a deeper understanding of the opportunities and challenges involved in introducing AI into HR processes. By applying both

frameworks, this study aims to identify not only the perceived benefits and drawbacks of AI but also the institutional and contextual factors that influence its successful adoption.

The Technology-Organization-Environment (TOE) framework, developed by Tornatzky and Fleischer, offers a comprehensive approach to understanding the multiple dimensions that affect technology adoption [30]. The first dimension, the technological context, focuses on the characteristics of the technology itself, such as its perceived usefulness, ease of use, complexity, and compatibility with existing systems [31]. In the case of AI in HRM, technologies that are seen as user-friendly, efficient, and aligned with current HR processes—such as tools for automating recruitment, managing employee records, or evaluating performance—are more likely to be accepted by HR professionals [32]. Conversely, if AI systems are perceived as difficult to use, unreliable, or poorly integrated with existing workflows, organizations may face resistance or hesitation during implementation [33]. The perceived value of the technology must therefore outweigh its perceived challenges. This technological perspective helps explain why some AI applications gain quick acceptance in HR functions while others encounter skepticism or rejection [34].

The second dimension of the TOE framework is the organizational context, which includes factors such as leadership support, resource availability, employee readiness, and organizational culture [35]. These internal factors can significantly influence how prepared an institution is to adopt and sustain new technologies [36]. In Tanzanian public institutions, for instance, strong support from top management is often essential for overcoming resistance to change and securing the necessary investments in training and infrastructure. The presence of skilled personnel who understand both HR processes and emerging technologies is also a key enabler of successful adoption. Furthermore, an open and adaptive organizational culture can encourage experimentation and innovation, whereas rigid bureaucratic systems may slow or even block the adoption of AI. Institutions that promote continuous learning and digital transformation are generally better positioned to integrate AI into their HR operations. Understanding these organizational dynamics is crucial for designing effective implementation strategies that align with the specific realities of the public sector.

1.2 Methodology

This study was conducted in three Local Government Authorities (LGAs) within Dar es Salaam, Tanzania: Ubungo, Kinondoni, and Temeke Municipal Councils. These councils operate under the Dar es Salaam City Council and were chosen because of their administrative significance and economic influence. As part of Tanzania's broader decentralization framework, LGAs are crucial to public service delivery and operate under the oversight of the Prime Minister's Office for Regional Administration and Local Government (PMO-RALG). They function according to the Local Government (District Authorities) Act, 1982, and the Local Government (Urban Authorities) Act, 1982 [37]. These institutions handle a wide range of responsibilities including service delivery, revenue collection, human capital development, and administrative governance. The selection of Kinondoni, Ubungo, and Temeke was therefore strategic, given their active engagement in public sector management and their relevance to the study's focus on AI integration within HRM.

The study adopted a cross-sectional research design to capture the current status of AI use in HRM practices across the selected councils. This design enabled a one-time, comprehensive data collection exercise, which is particularly useful for examining relationships between variables such as AI adoption, organizational preparedness, and perceived effectiveness in HRM functions. A positivist research philosophy guided the study, emphasizing objectivity, observable phenomena, and empirical analysis. The research relied on a quantitative approach, enabling structured data collection and statistical analysis, and providing a solid basis for identifying patterns, drawing generalizable conclusions, and minimizing potential researcher bias. This approach also allowed for comparisons between different HRM functions—such as recruitment, training, and performance management—in terms of how AI tools were understood and applied.

To ensure that participants were not only involved in HRM operations but also had a sound understanding of both HRM domains and AI-related concepts, specific eligibility and validation steps were taken before data collection. A total of 217 human resource practitioners participated in the study: 75 from Temeke, 72 from Kinondoni, and 70 from Ubungo. Participants included HR officers, department heads, directors, and supervisors, all of whom held functional responsibilities across key HRM areas, including recruitment and selection, employee development, performance appraisal, compensation, and succession planning. To validate their HRM expertise, the study only included practitioners with a minimum of two years of experience in public sector HRM roles. Their job titles and roles were verified through internal HRM records from the respective councils to confirm their practical engagement with HRM processes.

To ensure familiarity with AI-related concepts, a two-step validation process was implemented. First, participants completed a brief screening questionnaire before the main survey. This screening included basic questions assessing their awareness and understanding of AI applications in HRM, such as knowledge of AI-assisted recruitment tools, data analytics in

performance management, and automated payroll systems. Only those who demonstrated a minimum level of familiarity—defined as correctly identifying or describing at least three AI concepts relevant to HRM—were selected to proceed with the full survey. Second, during questionnaire administration, trained research assistants provided brief standardized explanations of key AI terms to reinforce understanding and clarify any ambiguities without leading responses. This ensured that all participants had a shared and sufficient conceptual grounding in AI, reducing the risk of misinterpretation.

The structured questionnaire was carefully designed to align with the Tanzanian public sector HRM context, using terminology and scenarios familiar to local practitioners. It included clear definitions and examples for more technical or emerging AI terms. The questionnaire was also pre-tested with a sample of HR practitioners not involved in the final study, allowing refinement of language to match respondent comprehension levels. Face-to-face administration further enabled clarification of any difficult questions in real time, enhancing the quality and accuracy of responses. These measures ensured that participants were well-informed and capable of providing meaningful input on both traditional HR domains and emerging AI technologies, strengthening the credibility of the data and the validity of the findings. Data collection took place between January 14 and April 15, 2023, through face-to-face administration to allow clarification of any ambiguous terms and to encourage accurate responses. This approach not only enhanced data quality but also ensured that participants understood the technological concepts being investigated, addressing the potential concern of uninformed responses.

Ethical considerations were strictly upheld to protect participants' rights and maintain research integrity. Approval for the study was secured from the President's Office for Regional and Local Government Authorities as well as the Dar es Salaam Regional Administrative Secretary. Participants were provided with written informed consent forms outlining the study's objectives, confidentiality assurances, and voluntary participation. They were informed that they could withdraw at any point without penalty. Data was securely stored and used strictly for academic purposes in compliance with national research ethics guidelines. These ethical practices helped foster trust and openness among respondents, contributing to the reliability and validity of the information gathered. A schematic overview of the research design and process is presented in Fig. 1, providing a visual representation of the methodology.

1.3 Risk categorization

In this study, risk levels were categorized based on the total number of "Agree" and "Strongly Agree" responses for each HRM component. This approach allows for a structured assessment of AI-related challenges in HRM. Components with 150 or more responses were classified as High Risk, indicating significant challenges that require immediate attention. Moderate Risk was assigned to components with responses between 140 and 149, reflecting notable but less critical concerns. Low Risk applied to components with responses between 130 and 139, suggesting minimal challenges. This classification provides a quantitative basis for understanding AI-related risks in HRM and it follows a method commonly used in technological research [37–39].

Therefore, the mathematical expression for the risk categorizations as used in study is given as:

$$R(N) = \begin{cases} HighRisk, N \geq 150 \\ ModerateRisk, 140 < N < 150 \\ LowRisk, 130 \leq N < 140 \end{cases}$$

Additionally, with indicator function that:

$$R(N) = H.1(N \geq 150) + M.1(140 \leq N < 150) + L.1(130 \leq N < 140)$$

whereas "H" represents High Risk, "M" represents Moderate Risk, "L" represents Low Risk, and $1(.)$ is an indicator function, returning 1 if the condition is met and 0 otherwise.

2 Results

Findings in Table 1 present an overview of the demographic and professional characteristics of human resource personnel across Temeke, Kinondoni, and Ubungo municipalities. The data reveals a slight male majority in all districts, with males comprising 53%–59% of the workforce. The age distribution indicates that the largest group is in the 31–40 age range, making up 35%–40% across the municipalities. Regarding education, most personnel hold a

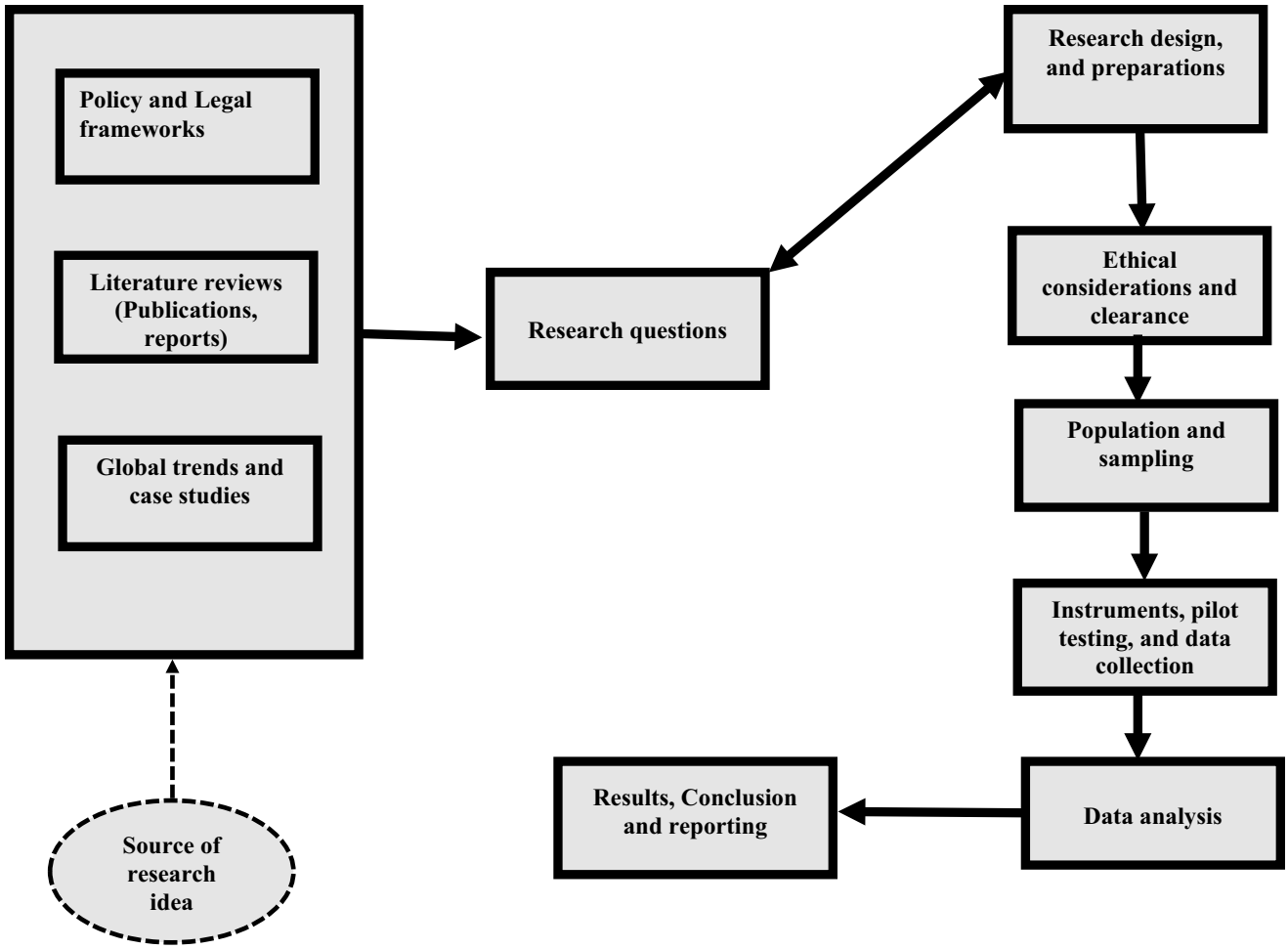

Fig. 1 Schematic diagram for the study methodology

bachelor's degree (53%–58%), followed by a smaller proportion with master's degrees (17%–25%). Work experience is predominantly in the 6–10-year range (40%–46%), suggesting a moderately experienced workforce across all districts.

Marital status trends show that the majority of HR personnel are married (56%–64%), with single, divorced, and widowed individuals representing smaller proportions. Job title distribution highlights HR Officer Level 2 as the most prevalent position in Temeke (40%) and Ubungo (50%), whereas Kinondoni displays a more even distribution of roles. Senior and Principal HR positions are less common, especially the latter, which accounts for only 11%–17% of roles. These findings provide a detailed demographic and professional profile of HR personnel, showcasing consistency in some attributes while highlighting district-specific differences.

The results in Table 2 reveal varying levels of understanding and awareness of AI and its applications in HR management among participants across the three municipalities. Most participants exhibit good general knowledge of AI, with Temeke (53%) and Kinondoni (49%) having the highest proportions, while Ubungo shows slightly higher rates of excellence (26%). When focusing on AI in HR, a majority still indicate good understanding, with Temeke leading (60%), followed by Kinondoni (56%) and Ubungo (50%). However, excellent comprehension is comparatively low, with Temeke at 13%, Kinondoni at 11%, and Ubungo at 17%, signaling a need for deeper, context-specific training. Basic or poor understanding persists among a minority, ranging from 7 to 33%, indicating a knowledge gap that could hinder effective AI utilization. Additionally, awareness of AI tools in HR management is predominantly moderate to high across all districts, with moderate awareness being the most prevalent in Kinondoni (42%) and Ubungo (40%). High awareness is most notable in Temeke (33%). Low or no awareness, while less common, still exists, particularly in Kinondoni (21%) and Ubungo (17%), indicating an opportunity for targeted efforts to improve familiarity with AI tools.

Table 1 General characteristics of human resource personnel across districts

Variable	Attribute	Temeke Municipal		Kinondoni Municipal		Ubungo Municipal	
		N	%	N	%	N	%
Sex	Male	44	59	38	53	40	57
	Female	31	41	34	47	30	43
Age	20–30 years	15	20	18	25	12	17
	31–40 years	30	40	25	35	28	40
	41–50 years	20	27	20	28	20	29
	Above 50 years	10	13	9	12	10	14
Education	Diploma	20	27	18	25	15	21
	Bachelor Degree	40	53	42	58	38	54
	Master Degree	15	20	12	17	17	25
Experience	1–5 years	25	33	20	28	18	26
	6–10 years	30	40	30	42	32	46
	Over 10 years	20	27	22	30	20	28
Marital Status	Single	20	27	18	25	15	21
	Married	45	60	40	56	45	64
	Divorced	5	7	10	14	6	9
	Widowed	5	7	4	6	4	6
Job Title	HR Officer Level 2	30	40	25	35	35	50
	HR Officer Level 1	10	13	15	21	12	17
	Senior HR	25	33	20	28	15	21
	Principal HR	10	13	12	17	8	11

Table 2 Participants'Perceptions of AI and Its Applications in HR Management

Variable	Attribute	Temeke Municipal		Kinondoni Municipal		Ubungo Municipal	
		N	%	N	%	N	%
General Understanding of AI	Excellent	15	20	12	17	18	26
	Good	40	53	35	49	30	43
	Basic	20	27	25	35	22	31
	Poor	5	7	8	11	10	14
Understanding of AI in HR	Excellent	10	13	8	11	12	17
	Good	45	60	40	56	35	50
	Basic	20	27	24	33	23	33
	Poor	5	7	5	7	10	14
Awareness of AI Tools in HR	High Awareness	25	33	20	28	22	31
	Moderate Awareness	35	47	30	42	28	40
	Low Awareness	10	13	15	21	12	17
	No Awareness	5	7	7	10	8	11

The results depicted in Fig. 2, a radar chart illustrating the perceived benefits and challenges of AI in HR management, highlight critical insights into participants'perceptions. Among the perceived benefits, increased efficiency stands out as the most recognized advantage, particularly in Ubungo (46%) and Temeke (40%). Improved decision-making follows, with Kinondoni reporting the highest acknowledgment (42%). Cost reduction is acknowledged by 25%–31% of participants, while enhanced recruitment is seen by a smaller proportion, ranging from 20% to 23%. Ease of using AI is ranked the lowest across all districts, with only 7%–9% of participants highlighting it as a benefit. This low ranking suggests a degree of skepticism or a lack of familiarity with the user-friendliness of AI technologies.

On the other hand, the challenges associated with AI adoption in HR reveal significant barriers that may hinder its effective implementation. The most pressing challenge is the lack of expertise, particularly in Kinondoni (35%) and

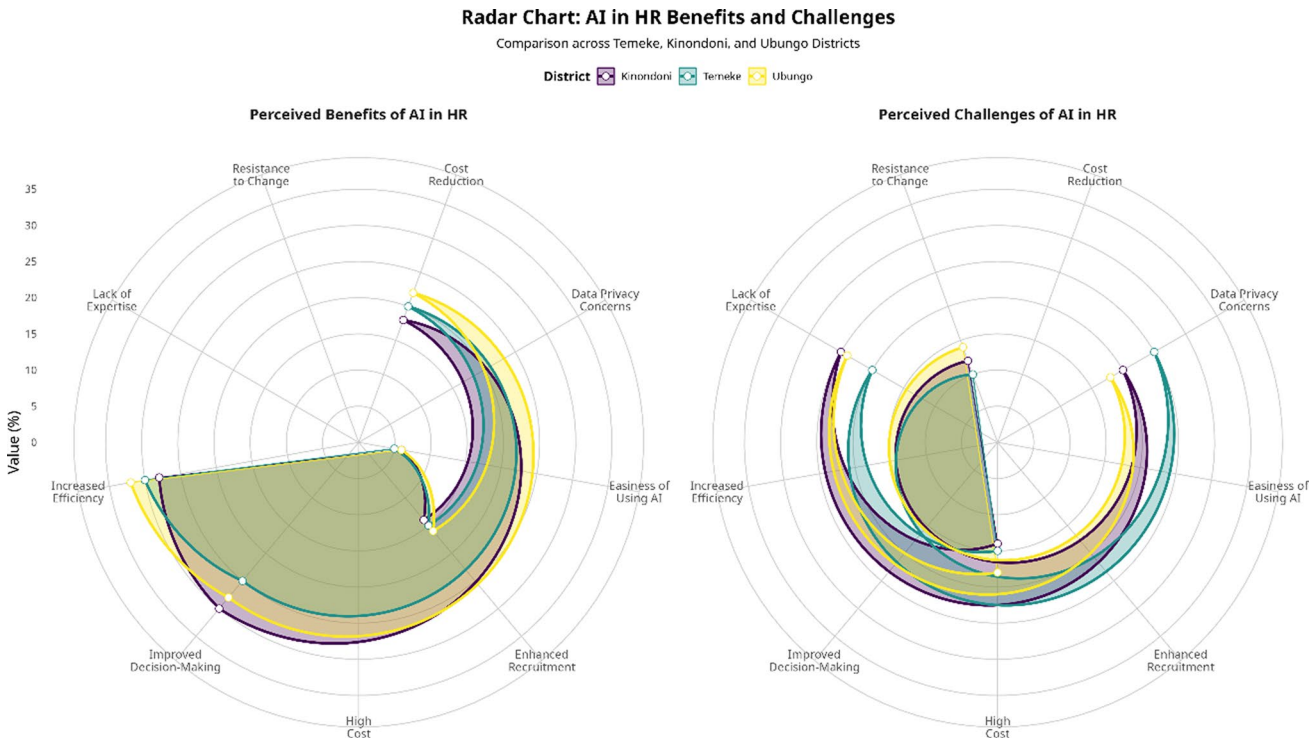

Fig. 2 Radar Chart on the Perceived Benefits and challenges of AI in HR Management

Ubungo (34%). Data privacy concerns are also prominent, with Temeke reporting the highest proportion of concern (33%). Resistance to change, though less significant, is observed in smaller proportions, ranging from 13% in Temeke to 20% in Ubungo. Additionally, the high costs of AI adoption emerge as a challenge, most notably in Ubungo (26%). These findings emphasize the dual reality of AI in HR management: while its benefits are recognized, challenges such as skill gaps, privacy concerns, and cost barriers must be addressed.

2.1 Benefits of artificial intelligence in human resource management

The results in Table 3 highlight the strong agreement among human resources personnel on the benefits of artificial intelligence (AI) across various HRM components. In workforce planning and recruitment, the majority agree that AI automates candidate screening efficiently (152 respondents, 70%) and enhances onboarding with personalized training modules for new hires (152 respondents, 70%). These findings emphasize AI's role in streamlining recruitment processes and improving employee integration.

In performance management, a significant proportion agree that AI generates actionable insights through performance tracking (148 respondents, 68%) and improves feedback accuracy through unbiased data analysis (148 respondents, 68%). Similarly, in training and development, 155 respondents (71%) agree that AI customizes employee training to address individual skill gaps, while 153 respondents (70%) acknowledge its role in identifying future skill requirements for strategic upskilling. These insights underline the recognition of AI's contribution to enhancing employee development and organizational preparedness.

For compensation and benefits, 155 respondents (71%) agree that AI uses predictive analytics to design competitive salary structures, while 153 respondents (71%) appreciate its ability to simplify benefit administration. In employee relations, 143 respondents (66%) agree that AI monitors employee sentiment effectively to detect dissatisfaction early, and 143 respondents (66%) endorse its capability to recommend proactive conflict resolution strategies. These findings showcase the growing trust in AI for improving employee relations and administrative efficiency.

In the domain of compliance and legal frameworks, 149 respondents (69%) agree that AI ensures adherence to regulatory changes through real-time alerts, and an equal proportion recognize its role in reducing human errors in compliance documentation; similarly, regarding workplace health and safety, 147 respondents (67%) confirm that AI effectively predicts and prevents safety risks by monitoring conditions, while 150 respondents (69%) acknowledge its utility in

Table 3 Benefits of artificial intelligence in human resource management

HRM Component	AI Benefits	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Workforce Planning and Recruitment	AI automates candidate screening to identify the best fit efficiently	87 (40%)	65 (30%)	43 (20%)	17 (8%)	5 (2%)
	AI enhances onboarding with personalized training modules for new hires	85 (39%)	67 (31%)	42 (19%)	18 (8%)	5 (2%)
Performance Management	AI generates actionable insights through continuous performance tracking	78 (36%)	70 (32%)	43 (20%)	19 (9%)	7 (3%)
	AI improves feedback accuracy with unbiased data analysis	80 (37%)	68 (31%)	44 (20%)	18 (8%)	7 (3%)
Training and Development	AI customizes employee training to address individual skill gaps	81 (37%)	74 (34%)	41 (19%)	15 (7%)	6 (3%)
	AI identifies future skill requirements for strategic upskilling	83 (38%)	70 (32%)	43 (20%)	15 (7%)	6 (3%)
Compensation and Benefits	AI uses predictive analytics to design competitive salary structures	83 (38%)	72 (33%)	39 (18%)	17 (8%)	6 (3%)
	AI simplifies benefit administration by automating complex processes	80 (37%)	73 (34%)	41 (19%)	17 (8%)	6 (3%)
Employee Relations	AI monitors employee sentiment to detect early signs of dissatisfaction	74 (34%)	69 (32%)	50 (23%)	19 (9%)	5 (2%)
	AI recommends proactive conflict resolution strategies	76 (35%)	67 (31%)	51 (24%)	18 (8%)	5 (2%)
Compliance and Legal Framework	AI ensures adherence to regulatory changes with real-time alerts	76 (35%)	73 (34%)	42 (19%)	20 (9%)	6 (3%)
	AI reduces human errors in compliance documentation and auditing	78 (36%)	71 (33%)	43 (20%)	19 (9%)	6 (3%)
Workplace Health and Safety	AI tracks workplace conditions to predict and prevent safety risks	79 (36%)	68 (31%)	44 (20%)	19 (9%)	7 (3%)
	AI facilitates wellness programs by monitoring employee health metrics	81 (37%)	69 (32%)	43 (20%)	17 (8%)	7 (3%)
HR Information Systems (HRIS)	AI streamlines data management and automates repetitive tasks in HR	85 (39%)	71 (33%)	41 (19%)	14 (6%)	6 (3%)
	AI provides predictive analytics for strategic workforce planning	82 (38%)	72 (33%)	42 (19%)	15 (7%)	6 (3%)
Succession Planning	AI identifies potential leaders through performance data analysis	84 (39%)	69 (32%)	44 (20%)	16 (7%)	4 (2%)
	AI maps career pathways to ensure seamless succession transitions	83 (38%)	68 (31%)	45 (21%)	17 (8%)	4 (2%)
Employee Engagement and Retention	AI predicts attrition risks by analysing engagement trends	82 (38%)	72 (33%)	43 (20%)	16 (7%)	4 (2%)
	AI designs personalized retention strategies using employee data insights	83 (38%)	71 (33%)	44 (20%)	15 (7%)	4 (2%)

enhancing wellness programs via health monitoring, underscoring AI's significant contributions to regulatory compliance and employee well-being. Additionally, within succession planning, 153 respondents (71%) affirm AI's effectiveness in identifying potential leaders through performance data analysis, and 151 respondents (69%) support its role in mapping clear career pathways for smoother transitions; meanwhile, in employee engagement and retention, 154 respondents (71%) recognize AI's accuracy in predicting attrition risks, and an equal proportion appreciate its capacity for crafting personalized retention strategies, collectively validating AI's extensive capability to enhance strategic human resource management and workforce stability across all districts.

The density plots in Fig. 3 illustrate the distribution of responses on the perceived benefits of AI in HR management across districts. The top left plot, representing all districts combined, shows a balanced density curve with moderate variability. The distribution suggests a central tendency where most respondents align closely in their perceptions of AI benefits, but there is still some degree of variability, reflecting differing levels of agreement across the combined dataset. For individual districts, Temeke shows a sharp peak, indicating strong consensus among respondents in this district regarding the benefits of AI. In contrast, Kinondoni displays a flatter density curve, suggesting a broader spread of responses and more varied opinions about AI benefits in HR management. Ubungo shows a slightly bimodal pattern, highlighting divided perceptions, where some respondents are strongly aligned with the benefits, while others differ in their agreement. These differences across districts point to varying levels of acceptance, understanding, or experience with AI's role in HR management.

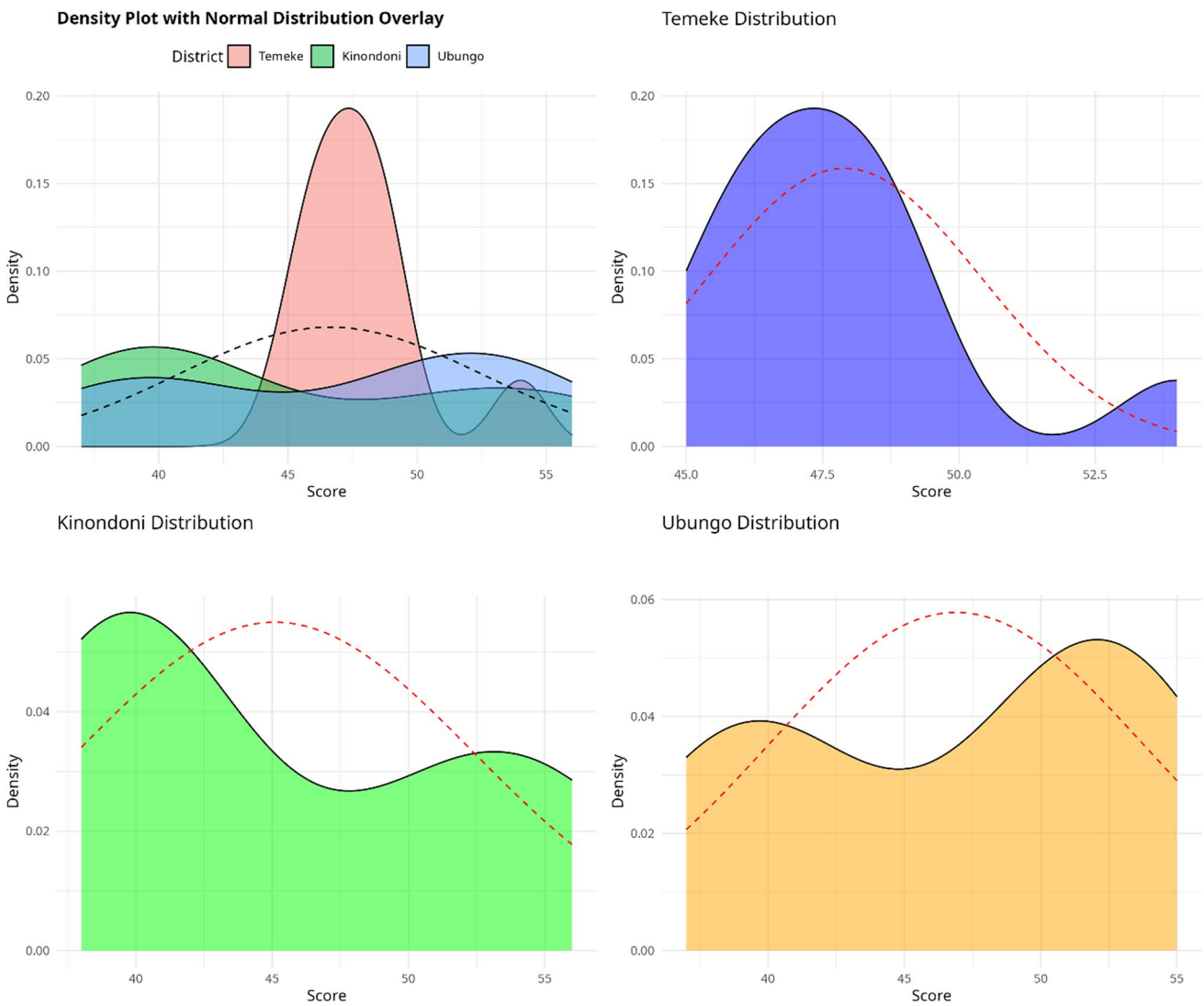

Fig. 3 Density plots with normal distribution overlay for the benefits of AI across districts

Moreover, results in Table 4 indicate that Temeke has the highest mean score (47.9) with the lowest variability (SD = 2.51), a positive skewness (1.400), and high kurtosis (4.589), indicating concentrated and consistent responses, though slightly skewed toward higher scores. Kinondoni shows the lowest mean (45.1) and the highest variability (SD = 7.24), with a moderate positive skewness (0.500) and kurtosis (1.481), reflecting more diverse responses. Ubungo has a balanced mean (46.9), moderate variability (SD = 6.90), and a slight negative skewness (−0.275), indicating a more even distribution of scores leaning slightly toward lower values. These results suggest that while all districts perceive AI benefits, Temeke shows greater consistency, Kinondoni has more variability, and Ubungo exhibits balanced perceptions.

2.2 Challenges of artificial intelligence in human resource management

The results in Table 5 highlight several perceived challenges associated with using AI across different HRM components. In workforce planning and recruitment, a majority of respondents (72%) agree that AI might unintentionally perpetuate biases in hiring and that automated recruitment processes can feel impersonal. This underline concerns about fairness and the human touch in AI-driven recruitment processes. A notable proportion (18%–19%) strongly disagrees, suggesting varying levels of trust in AI systems for these tasks.

In performance management, 69% of respondents agree that AI may overly emphasize measurable aspects, ignoring qualitative factors, and that continuous monitoring by AI raises privacy concerns. These findings suggest that while AI is seen as a useful tool, there is discomfort about its focus on quantifiable metrics and potential invasion of employee privacy. Similarly, in training and development, 69% agree that AI recommendations may misalign with organizational goals or individual needs, and the high cost of integration is seen as a barrier. This indicates a need for better alignment between AI capabilities and organizational strategies.

For compensation and benefits, 72% of respondents believe that AI-driven decisions about salaries and benefits may lack transparency, and that such systems might struggle to address unique employee situations. Transparency and adaptability are key concerns in using AI for compensation-related tasks. In employee relations, 67% agree that AI cannot handle the emotional nuances of workplace conflicts, and employees may distrust AI-driven systems for conflict resolution.

In compliance and legal frameworks, 70% of respondents agree that AI might misinterpret complex regulations, leading to non-compliance, and that balancing efficiency with ethical considerations is a challenge. Similar concerns are evident in workplace health and safety, where 69% agree that AI might misidentify safety risks or raise privacy concerns with health data collection. Finally, in succession planning and employee engagement, respondents highlight challenges such as overlooking intangible leadership qualities (68% agree) and failing to address diverse employee needs (69% agree). These findings suggest that while AI offers significant potential, its limitations in understanding in-depth, ethical, and emotional aspects require complementary human oversight to maximize its effectiveness.

2.3 Levels of risks of AI in human resource management

Understanding the challenges of AI in HRM without assessing the associated risks provides an incomplete picture and limits actionable insights. Figure 4 illustrates the risk levels associated with AI implementation across various HRM components, offering valuable insights into the challenges in each area. Components classified as High Risk, including Human Resource Information Systems (HRIS), Compensation and Benefits, Training and Development, Employee Engagement and Retention, Workforce Planning and Recruitment, and Performance Management, face significant challenges. While AI presents opportunities in these areas, high-risk classification indicates persistent concerns such as bias, lack of transparency, and implementation complexity. Addressing these risks is crucial, as failure to do so could undermine AI's effectiveness, leading to inefficiencies and employee dissatisfaction.

Components classified as Moderate Risk, such as Succession Planning and Compliance and Legal Framework, reflect notable but less severe challenges. These areas may have some barriers, such as balancing AI efficiency with ethical considerations

Table 4 Normality test on perceived benefits of AI across districts

District	Mean	SD	Shapiro P Value	Skewness	Kurtosis
Temeke	47.9	2.51	0.059	1.400	4.589
Kinondoni	45.1	7.24	0.021	0.500	1.481
Ubungo	46.9	6.90	0.097	−0.275	1.384

Table 5 Challenges of artificial intelligence in the human resource management

HRM Component	AI Challenges	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Workforce Planning and Recruitment	AI may unintentionally perpetuate biases in hiring decisions	80 (37%)	75 (35%)	7 (3%)	20 (9%)	40 (18%)
	Automated recruitment processes can feel impersonal to candidates	77 (36%)	73 (34%)	7 (3%)	18 (8%)	42 (19%)
Performance Management	AI may overly emphasize measurable aspects, ignoring qualitative factors	78 (36%)	72 (33%)	5 (2%)	19 (9%)	43 (20%)
	Continuous monitoring by AI tools raises privacy concerns among employees	82 (38%)	68 (31%)	6 (3%)	21 (10%)	40 (18%)
Training and Development	AI recommendations may not align with organizational goals or individual needs	81 (37%)	70 (32%)	7 (3%)	17 (8%)	42 (19%)
	High costs of integrating AI into training systems can be a barrier	79 (36%)	72 (33%)	6 (3%)	19 (9%)	41 (19%)
Compensation and Benefits	AI-driven decisions about salary and benefits may lack transparency	83 (38%)	74 (34%)	4 (2%)	17 (8%)	39 (18%)
	AI systems may fail to address unique or non-standard employee situations	80 (37%)	70 (32%)	5 (2%)	19 (9%)	43 (20%)
Employee Relations	AI tools are unable to handle emotional nuances of workplace conflicts	76 (35%)	69 (32%)	8 (4%)	20 (9%)	44 (20%)
	Employees may distrust AI-driven systems for conflict resolution	75 (35%)	71 (33%)	7 (3%)	21 (10%)	42 (19%)
Compliance and Legal Framework	AI might misinterpret complex regulatory requirements, leading to non-compliance	78 (36%)	73 (34%)	5 (2%)	21 (10%)	40 (18%)
	Balancing AI efficiency with ethical considerations is challenging	80 (37%)	71 (33%)	6 (3%)	19 (9%)	41 (19%)
Workplace Health and Safety	AI might misidentify workplace safety risks, causing false positives or negatives	79 (36%)	70 (32%)	5 (2%)	20 (9%)	43 (20%)
	Collecting health data for AI raises privacy and data protection concerns	81 (37%)	68 (31%)	5 (2%)	19 (9%)	44 (20%)
HR Information Systems (HRIS)	AI may face integration challenges with existing HR systems	82 (38%)	71 (33%)	7 (3%)	17 (8%)	40 (18%)
	Managing large data volumes generated by AI can overwhelm HR departments	81 (37%)	72 (33%)	5 (2%)	18 (8%)	41 (19%)
Succession Planning	AI may overlook intangible qualities like leadership potential and cultural fit	79 (36%)	69 (32%)	6 (3%)	20 (9%)	43 (20%)
	Historical data used by AI might hinder diversity in leadership	78 (36%)	70 (32%)	7 (3%)	18 (8%)	44 (20%)
Employee Engagement and Retention	AI lacks emotional understanding, impacting its ability to address engagement challenges	81 (37%)	72 (33%)	5 (2%)	18 (8%)	41 (19%)
	AI-driven retention strategies may not account for diverse employee needs	80 (37%)	70 (32%)	6 (3%)	19 (9%)	42 (19%)

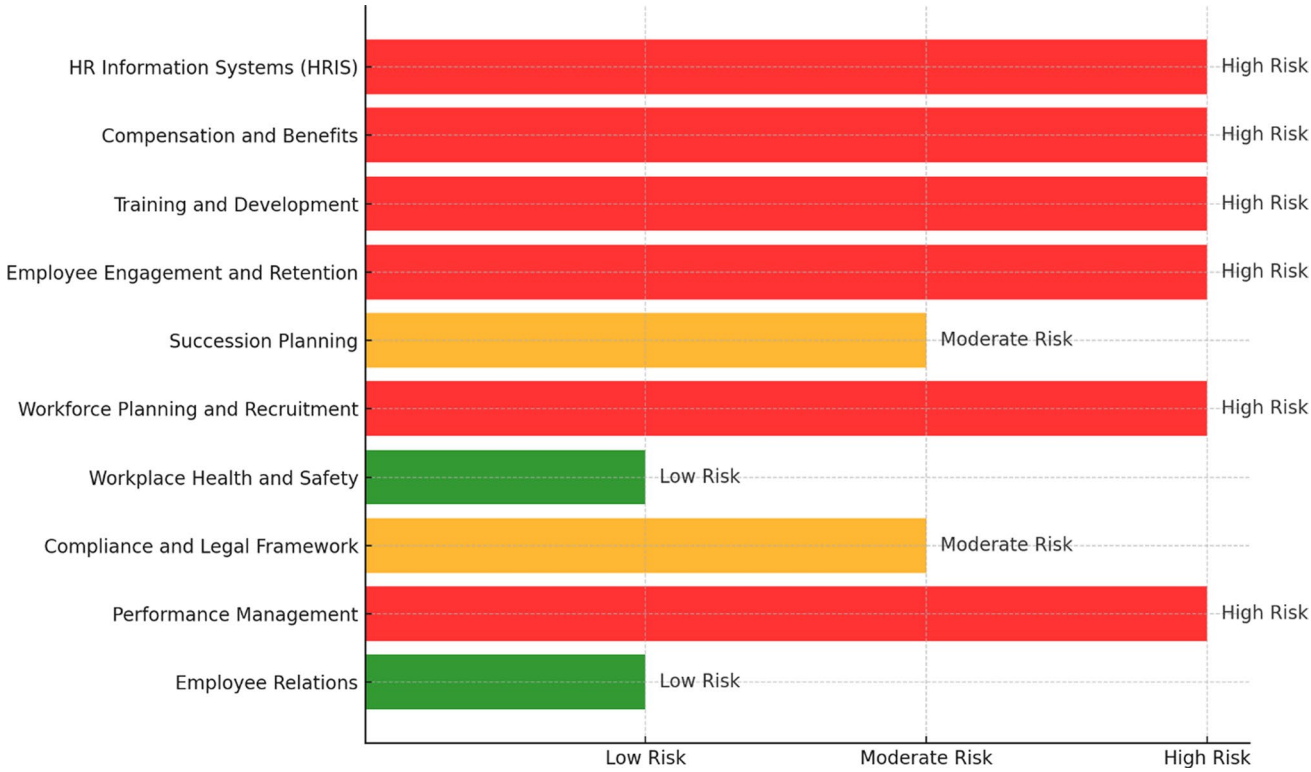

Fig. 4 Risks associated with AI on Human Resource Management

or ensuring leadership diversity, but the overall risk level is manageable with targeted improvements. Moderate risk indicates the need for proactive but less urgent interventions compared to high-risk components.

Low Risk components, including Workplace Health and Safety and Employee Relations, show minimal challenges, suggesting that AI is effectively implemented in these areas with fewer obstacles. Low risk indicates that these components benefit from well-established processes or minimal perceived drawbacks, making them relatively stable. This categorization means resources can be allocated to higher-risk areas while continuing to monitor and sustain the strengths in low-risk components. Generally, the classification provides a roadmap for prioritizing resources and efforts to optimize AI implementation in HRM.

Table 6 presents the results of the Chi-Square test on various HRM components and their associated risk levels. The results indicate that certain HRM components, such as Human Resource Information Systems (HRIS), Workforce Planning and Recruitment, and others with p-values of 0.005, 0.004, and 0.019 respectively, show strong statistical significance, as their p-values are below 0.01 and 0.05. This means that there is a very low probability that the observed differences in risk levels for these components are due to chance, thus confirming that these components are significantly associated with their risk levels. For components such as HRIS and Workforce Planning and Recruitment, this significance at the 1% level indicates that their risk levels are highly influenced by the factors being studied, and interventions may be crucial in these areas. Other components like Compensation & Benefits, Training and Development, Employee Engagement and Retention, Performance Management, Succession Planning, and Compliance and Legal Framework also exhibit significant results, with p-values below 0.05 (5%) or 0.1 (10%), which suggests that these components are statistically significantly associated with risk levels, though with varying degrees of strength. The significance for Workplace Health and Safety indicates a weak but still noticeable relationship between this component and its risk level. When a component is found to be "not significant" it means there is no strong statistical evidence to conclude that the component's risk level is directly influenced by the factors being tested.

Table 6 Chi square test on the HRM components and risk levels

HRM component	Risk Level	Chi-Square (χ^2)	Critical Value (df = 2)	P values	Significance
Human Resource Information Systems (HRIS)	High	34.62	10.151	0.005	***
Compensation and Benefits	High	15.63	10.151	0.041	**
Training and Development	High	19.39	10.151	0.035	**
Employee Engagement and Retention	High	14.03	10.151	0.039	**
Workforce Planning and Recruitment	High	29.05	10.151	0.004	***
Performance Management	High	16.88	10.151	0.019	**
Succession Planning	Moderate	15.55	10.151	0.045	**
Compliance and Legal Framework	Moderate	17.01	10.151	0.020	**
Workplace Health and Safety	Low	11.07	10.151		*
Employee Relations	Low	15.85	10.151		**

*p < 0.1 (10%), **p < 0.05 (5%), ***p < 0.01 (1%)

3 Discussion

This study highlights the dual nature of AI in HRM, emphasizing both its transformative benefits and inherent challenges. The findings align with previous research indicating that AI significantly enhances efficiency in HR processes, such as recruitment, training, and decision-making [40–43]. However, the variability in perceptions across districts suggests uneven implementation and adoption, which could limit AI's potential. For example, Temeke's consistent results imply higher levels of readiness and alignment with AI tools, while Kinondoni's variability underscores the need for better capacity-building and resource allocation [44, 45].

The strong agreement on AI's ability to improve recruitment and training processes reflects its potential to revolutionize HRM by automating repetitive tasks and enabling data-driven decisions. These results are consistent with studies demonstrating that AI enhances candidate screening and personalized training programs, reducing time and costs [46–49]. However, concerns about impersonal interactions and biases, as highlighted in this study, mirror findings by Zanke et al., [50], who argue that AI can perpetuate systemic inequalities if not carefully managed. This suggests that while AI has benefits, its implementation must be accompanied by strategies to ensure fairness and inclusivity.

Privacy concerns and the inability of AI to address qualitative aspects of HR processes, such as emotional nuances and ethical considerations, emerged as significant challenges. Similar issues have been raised in other studies, which warn that reliance on AI for performance management can lead to employee distrust and resistance [51–53]. Continuous monitoring by AI tools, while efficient, risks creating a surveillance culture, undermining employee morale and productivity [54, 55]. These findings suggest the need for a balance between AI-driven efficiency and maintaining human-centric approaches to HR.

The High-Risk categorization of HR Information Systems (HRIS), Compensation and Benefits, and Performance Management highlights the critical challenges these areas face. Other studies support these findings, indicating that transparency and accountability in AI-driven decisions remain pressing issues [56, 57]. For example, salary recommendations and performance evaluations driven by algorithms can lack clarity, eroding trust among employees [58, 59]. Addressing these risks requires robust ethical frameworks and clear communication about AI's decision-making processes. Additionally, moderate risk components, such as Succession Planning and Compliance and Legal Framework, reflect areas where challenges are less severe but still require attention. Previous research suggests that AI can inadvertently reinforce traditional biases in leadership selection and struggle with interpreting complex regulatory requirements [60–64]. These issues highlight the importance of integrating human oversight and ethical safeguards into AI systems to ensure they align with institutional goals and values.

Low Risk components, such as Workplace Health and Safety and Employee Relations, suggest areas where AI implementation has been more effective. Studies have shown that AI can successfully identify workplace safety risks and facilitate data-driven wellness programs, as supported by these findings [65, 68–70]. However, the study also cautions that even in low-risk areas, there is room for improvement, particularly regarding privacy concerns related

to health data collection. These results suggest that while these components may be performing well, continuous monitoring and refinement are necessary to sustain their success.

Additionally, while this study highlights AI's potential to transform HRM in public institutions, it also uncovers significant challenges that align with findings in existing literature. For instance, the study by Votto et al. [66] demonstrates the usefulness of AI in HR-related practices such as decision-making, resource utilization, and problem-solving. However, it also reveals a lack of transparency in how AI is involved in these decisions, which raises concerns about the fairness and accountability of AI-driven processes. Effective AI adoption, therefore, requires addressing biases, ensuring transparency, and striking a balance between efficiency and ethical considerations.

Moreover, the literature questions the effectiveness of AI in human resource management. Alhamoud and Al-Kasasbeh [67] highlight a problem related to data science, particularly in HR outcomes that are rare or happen infrequently, such as dismissals. Techniques like machine learning and other data science methods typically require large datasets to produce accurate results. Public institutions must prioritize capacity-building, employee engagement, and policy development to maximize AI's benefits while minimizing its risks [71–73]. These findings provide a roadmap for leveraging AI in HRM, ensuring equitable and ethical implementation in line with recommendations from recent studies.

Also, the results from the Chi-Square test reveal varying levels of statistical significance across different HRM components, emphasizing the complexity of AI integration within HR practices. Significant components such as HRIS, Workforce Planning & Recruitment, and others show that certain HRM functions are more prone to challenges that impact their risk levels, which aligns with previous studies highlighting AI's transformative yet challenging nature in HRM [74, 75]. The findings underscore that areas like HRIS and Workforce Planning are highly sensitive to the risk factors associated with AI adoption, which may require targeted interventions to mitigate risks and enhance AI effectiveness [76, 77]. On the other hand, components like Employee Relations and Workplace Health & Safety, while still significant to some degree, indicate that AI's impact on these areas may be less pronounced, suggesting that these components may benefit from less intensive AI intervention [78]. The results also point to the need for institutions to develop a nuanced, context-specific approach to AI integration, as the risk levels differ across HRM functions, highlighting the importance of ethical considerations, privacy, and transparency in AI's implementation [79, 80].

4 Conclusion

This study underlines the transformative potential of AI in human resource management within public institutions, highlighting its ability to streamline processes, enhance decision-making, and personalize employee development. However, the findings also reveal significant challenges, such as biases, lack of transparency, privacy concerns, and the inability to address qualitative aspects of HR processes. These challenges vary across districts, with Temeke demonstrating greater consistency in AI adoption, while Kinondoni and Ubungo face more diverse or balanced perceptions. These disparities suggest that while AI holds promise, its implementation must be carefully managed to ensure equitable and effective outcomes.

To address these challenges, public institutions must adopt a multifaceted, practical, and evidence-based approach. First, training programs should be designed based on the specific AI applications used in HRM, incorporating real-case scenarios of AI-related risks such as biased hiring decisions or privacy breaches. Providing HR professionals with hands-on training in AI ethics, data handling, and algorithmic auditing will enhance their ability to detect and mitigate biases. Second, institutions should establish clear ethical guidelines and regulatory frameworks, drawing insights from successful AI governance models in other sectors. For example, integrating explainable AI (XAI) principles into HR systems can enhance transparency, ensuring that employees understand how AI-driven decisions are made. Third, human oversight should not be symbolic but integrated meaningfully—HR professionals should have authority to override AI decisions when ethical concerns arise, ensuring AI supports rather than replaces human judgment in critical areas like performance evaluations and disciplinary actions.

Lastly, public institutions, particularly those in Tanzania's Temeke, Ubungo, and Kinondoni Municipal Councils, should invest in AI solutions that align with their workforce needs and infrastructure capacity. Instead of generic AI tools, institutions should pilot AI applications in specific HR functions—such as using predictive analytics to anticipate workforce shortages—before scaling up. Lessons from low-risk AI implementations, such as workplace safety monitoring, should be adapted to improve high-risk areas like HR Information Systems and Compensation and Benefits. By adopting a phased and evidence-based approach, public institutions can maximize AI's benefits, address its challenges, and create a balanced, effective, and ethical framework for modernizing HR practices.

4.1 Implications of the study

The findings of this study emphasize the need for public institutions to develop clear governance frameworks that address biases, transparency issues, and privacy concerns in AI-driven HRM processes. AI has the potential to enhance efficiency in HR functions such as recruitment, training, and performance evaluation, but without proper ethical guidelines, there is a risk of unfair decision-making and discrimination. Institutions should establish policies that regulate AI's role in HRM, ensuring that automated decisions remain transparent and accountable. Additionally, AI's ability to process large amounts of HR data can improve workforce planning and decision-making; however, the study highlights the importance of balancing AI with human oversight. While AI can streamline HR tasks, human intelligence remains crucial in handling sensitive issues such as conflict resolution, employee well-being, and ethical decision-making. A hybrid approach that integrates AI and human expertise will ensure that AI is used as a supportive tool rather than a sole decision-maker.

Another key implication of this study is the need for a context-specific approach to AI implementation. The findings reveal disparities in AI adoption across different districts, indicating that institutions should assess their readiness, resource availability, and technological infrastructure before integrating AI into HRM. Public organizations with well-developed digital HR systems may experience smoother AI implementation, while those lacking resources may face challenges in adoption. Institutions must invest in capacity-building initiatives to equip HR professionals with the necessary skills to manage AI tools effectively. Additionally, ethical AI adoption should be accompanied by continuous monitoring and evaluation to identify potential risks and ensure compliance with organizational values. By addressing these issues, public institutions can harness AI's benefits while minimizing risks, leading to more efficient, transparent, and equitable HRM practices.

4.2 Limitations of the study

While this study provides valuable insights into AI integration in HRM within public institutions, it has some limitations that should be acknowledged. The research focused on three municipal councils in Tanzania—Temeke, Kinondoni, and Ubungo—which may not fully represent all public institutions globally. Differences in economic development, institutional policies, and technological infrastructure across countries mean that the findings may not be directly applicable to other contexts. Additionally, even within Tanzania, the study was limited to local government authorities, excluding other public institutions such as ministries, agencies, and state-owned enterprises. This may limit the generalizability of the findings to public institutions with different operational structures and levels of AI adoption. Furthermore, the study employed a cross-sectional research design, which captures data at a single point in time and does not account for changes or trends over time. A longitudinal study could provide deeper insights into how AI adoption evolves and its long-term impacts on HRM.

Another limitation is that this study focused exclusively on public institutions, thereby overlooking AI adoption in private organizations where technological advancements may be more rapid and implementation challenges may differ. Future research should consider comparative studies between public and private institutions to gain a broader understanding of AI's impact on HRM. Additionally, this study did not explore the legal and regulatory frameworks that govern AI adoption in HR processes. Understanding how existing labor laws facilitate or hinder AI integration is essential for ensuring compliance and ethical use. Moreover, the study relied on descriptive statistical analyses without employing rigorous econometric methods to establish causal relationships between AI adoption and HRM outcomes. Future studies could incorporate advanced econometric techniques to analyze the direct and indirect effects of AI on HR performance, employee satisfaction, and organizational efficiency. Addressing these limitations in future research will provide a more comprehensive perspective on AI adoption in HRM.

4.3 Future research

To build on the findings of this study, future research should expand the scope to include a broader range of public institutions, both within Tanzania and internationally, to enhance the generalizability of the results. A comparative study between public and private institutions would also be beneficial in identifying best practices, challenges, and

differences in AI adoption. Additionally, examining the role of regulatory and legal frameworks in AI-driven HRM would provide insights into how policies shape AI implementation, helping organizations navigate ethical and legal challenges.

Another important area for future research is the exploration of specific AI technologies in HRM. This study examined AI adoption broadly, without differentiating between machine learning, deep learning, or natural language processing. Future research should investigate how different AI techniques impact various HR functions, such as talent acquisition, performance management, and employee engagement. Understanding the strengths and limitations of specific AI tools will enable organizations to make informed decisions about their adoption and implementation strategies. By addressing these gaps, future studies can contribute to a more comprehensive understanding of AI's role in transforming HRM in both public and private institutions.

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Data availability Data will be available upon reasonable request based on the guideline provided by the ethical approval authority.

Declarations

Ethics approval and consent to participate. a) The study was performed in line with the principles of the Declaration of Helsinki. Ethical approval for this study was granted by the President's Office Regional and Local Government Authorities. b) Authors declare that all methods were carried out in accordance with relevant guidelines and regulations. c) Authors declare that a written consent was obtained from all participants.

Competing interests The authors declare no competing interests.

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